

Numberes Cooperative

- **Location / Counties served:** Baringo County
- **Contact / Date recorded:** Isaac (Chairman), 07 22 99 3512; 22/09/25
- **Area:** 12,800 acres (≈5,120 ha)
- **Primary enterprises:** Mixed cropping and livestock; maize dominant, potatoes, dairy, vegetables, Napier for feed.
- **Crop mix & calendar:** Maize ~60% (plant Mar–Oct); Potatoes ~planted Feb, Jun, Sep (three times/year); Green peas, carrots, cabbage, kale, tomato, avocado, Napier grass.
- **Yields & prices noted:** Potatoes 60 × 50 kg bags/acre; maize 20 × 90 kg bags/acre; green peas KES 60/kg; carrots 450/20 kg.
- **Soil & inputs:** 100% fertilized area using ammonium, phosphate, NPK; organic manure used; all rainfed; no irrigation; no biochar; planting cost/acre: maize KES 3,300; potatoes KES 60,000.
- **Tillage & residue:** Disk ploughing (50% tractor or ox); previously burned but stopped; monocropping.
- **Livestock:** Grazing: 80% free-range, 15% confined, 5% zero-grazing; cattle 40%, sheep 10%; wet-season feed = Napier, dry-season = hay.
- **Livestock risks & market issues:** Feed scarcity, lack of improved breeds, poor housing; frequent fluctuation in feed supplement and milk prices.
- **Strengths:** Large land area, diversified cropping, integrated fodder (Napier).
- **Key challenges / priority actions:** Introduce improved breeds and housing, secure feed through dry-season forage or fodder conservation, explore irrigation or water-harvesting to reduce risk.

Lanyuk Cooperative

- **Location / Counties served:** Narok County
- **Project Area:** 37.2 km perimeter/ 4,636 ha area
- **Contact / Date recorded:** Joseph Ngugi (Chairman); 27/09/25
- **Primary enterprises & area share:** Potatoes 40%, corn 20%, wheat 15%, barley 15%, carrots 3%, cabbage 3%, peas 2%, canola 2% (canola is new).
- **Cropping calendar:** Wheat, barley, canola July–Jan (some indicate Mar–Dec); potatoes planted year-round.
- **Yields & production metrics:** Potatoes 50 t/acre; potatoes avg ~1,800 × 80 kg bags; corn 1.5–2 t/acre; wheat/barley 1.3 t/acre; cabbage 5 t/acre; carrots 10 t/acre; peas 3 t/acre; canola 2 t/acre.
- **Soil & inputs:** Noted fertilization status not specified; rainfed.
- **Strengths:** High potato productivity and year-round potato planting; diversified cereal mix including new canola.
- **Key challenges / priority actions:** Validate input use and soil fertility management for cereals; scale canola best-practices; improve storage and market linkages for large potato volumes.

Kabianga Cooperative

- **Location / Counties served:** Kericho County (subcounties: Bureti/Bareti, Belgut, Ainamoi, Soin)
- **Project Area:** 33.2 miles/53.43022 km perimeter and 59.3 sq miles/ 15358.63 hectares area
- **Contact / Date recorded:** Franklin (Manager); 25/09/25

- **Membership & land use:** ~5,000 farmers registered, ~2,900 active; tea dominates (~80% historically, declining); some farmers converting tea to pasture (30–40% converted; ~25% farms reduced to grazing post-2021).
- **Primary enterprises:** Tea (primary), maize, avocados, potatoes, cabbage, bananas, livestock (increasing).
- **Tea production:** Harvest every 10 days (~3 times/month); average yield 600–700 kg/acre/month; average price KES 20/kg.
- **Soil & inputs:** 100% of area receives fertilizer; synthetic NPK, phosphorus, potassium; manure used for subsistence; two fertilizer applications per year; synthetic fertilizer application noted as 4 × 50 kg bags/acre twice/year.
- **Water & tillage:** All rainfed; tillage manual and tractor ploughing; till to 2 ft depth; 20% soil area disturbed; crop residue removed.
- **Livestock & land change:** Mostly zero-grazing; some expansion of milk collection area and mapping; farmers converted tea to cows during COVID period.
- **Risks & constraints:** Feed scarcity, diseases, market access, lack of improved breeds; minimal liming for avocados.
- **Strengths:** Large membership base, established tea processing/harvesting rhythm, full fertilizer coverage.
- **Key challenges / priority actions:** Monitor land-use change from tea to pasture, quantify manure rates and soil health, deploy pasture/feed solutions, strengthen market access for milk and diversify value-addition.

Kipsigis Cooperative

- **Location / Counties served:** Bomet / Kericho
- **Project Area:** 44km perimeter, 9,678 ha
- **Contact / Date recorded:** Geoffrey (CEO); 26/09/25
- **Membership & scale:** 5,500 members.
- **Primary enterprises & outputs:** Eucalyptus and tea production; harvest volume noted 18,000 m3 (eucalyptus and tea combined).
- **Harvesting rhythm & management:** Manual tea harvest every 10 days; mechanical harvesting every 21 days for some operations.
- **Soil & inputs:** No organic manure or compost used; NPK applied 100–250 kg/ha by hand across 100% of area.
- **Tillage & soil practices:** No tillage; no liming; replant/replace after ~50 years.
- **Strengths:** Large membership and significant volume; consistent fertilizer application.
- **Key challenges / priority actions:** Introduce organic matter / compost to improve soil health, evaluate liming where soils acidify, assess sustainable harvest rotation and replanting planning.

Adu Cooperative

- **Location / Counties served:** Kilifi County
- **Project Area:** 53.3 km perimeter/ 8,984 ha area
- **Contact / Date recorded:** Chairman; 02/10/25
- **Membership & origins:** Started 2020 with 9 men and 8 women; now 285 members, 188 women, inclusive of people with disabilities and youth.
- **Area & land use:** Estimated 1,800 acres total, 900 acres actively farmed; 75% subsistence.

- **Primary enterprises & production:** Corn (50% of harvest), cowpeas (25%), green peas (25%); corn yields ~15–20 bags/acre (1 bag = 90 kg); corn price KES 3,000/bag; cowpeas KES 8,000/bag (2 bags/0.5 acre); green peas KES 12,000/bag (2 bags/0.5 acre).
- **Livestock & dairy:** Daily milk avg 180 L/day; milk sold to nearby schools and locals; leftover milk used for yogurt value addition. County supplied milk cans and spray pumps; received 17 hay bales/files.
- **Soil & inputs:** 75% of area fertilized; organic manure used (manure rate ~100 kg/acre); no synthetic fertilizer used due to cost and soil concerns; sandy soils.
- **Tillage & residue:** Tillage manual and animal-drawn; plough depths: 6 inches with cows, 8–12 inches with tractor (tractor used ~70%); 100% soil area disturbed; crop residue cut and fed to cows; peas left on fields. Burning stopped.
- **Strengths:** Rapid membership growth, strong women's participation, active value-addition (yogurt), clear local milk markets.
- **Key challenges / priority actions:** Improve soil fertility on sandy soils (targeted organic matter, mulching, composting), secure synthetic fertilizer alternatives or subsidies if needed, scale fodder conservation to increase dry-season milk production.

Eor-Emayian Cooperative

- **Location / Counties served:** Eor Emayian area (contact: Charles, Chairman)
- **Project Area:** 29.2 km perimeter/ 4,387 hectares
- **Partners / Notes:** Gazelle / Agricentric involvement; 300 farmers.
- **Primary enterprises:** Mixed cropping with dairy presence; maize 40%, potatoes 60%, beans, cabbage, kale, avocados; limited intercropping (~5% seasonally).
- **Soil & inputs:** Farmers report acidifying soils and falling yields; combined use of chemical and organic fertilizers is increasing; practice includes drying and mixing manure with chemical fertilizer. Organic-only use ~5%, mixed organic+chemical ~5%, chemical fertilizer ~90% of farmers. Common chemical inputs: DAP widely used (~90%), NPK (~10%), CAN; spraying for application. Rabbit urine used by some as combined fertilizer + pesticide.
- **Irrigation & residues:** Rainfed; residues mostly fed to cows; ~5% burn residues though discouraged; potato residues often left on fields.
- **Liming & soil remediation:** No liming practiced.
- **Strengths:** Clear crop focus on potatoes and maize, experimentation with integrated organic/chemical approaches, existing dairy integration for residue reuse.
- **Key challenges / priority actions:** Address soil acidity with liming and soil testing, increase organic matter (composting at scale), formalize integrated nutrient management, monitor yields and plan targeted soil health interventions.

Summary recommendation (across cooperatives)

Total project hectares: 48,163 ha

- **Soil health:** Prioritize soil testing, liming where acidic, and scaling compost/organic matter to reduce dependence on synthetic fertilizer and restore yields.
- **Feed & livestock resilience:** Scale fodder conservation (hay, silage), introduce improved breeds and better housing, and formalize dry-season feed supply.
- **Water & irrigation:** Investigate low-cost water-harvesting or small-scale irrigation in high-risk sites to reduce rainfed vulnerability.

- **Value addition & markets:** Strengthen milk value-addition (yogurt), storage and potato market channels, and direct buyer linkages to reduce price volatility.
- **Data & planning:** Standardize simple metrics (yield per acre, fertilizer rates, livestock numbers) and maintain maps of collection zones to manage expansion and services.